Web Dev Mastery

📘 Topics Covered:

1. CSS Overflow
2. CSS Object-Fit
3. Object-Fit: Cover

🔹 1. CSS Overflow

✅ When to Use:

- To **control what happens** when content overflows its container.
- Use it in scrollable sections, modals, cards, or fixed-height containers.

🧠 Explanation:

The `overflow` property determines how to handle content that exceeds the container's dimensions.

css

```
overflow: visible | hidden | scroll | auto;
```

- `visible`: default; content is not clipped.
- `hidden`: extra content is hidden.
- `scroll`: adds scrollbars even if not needed.
- `auto`: adds scrollbars only when needed.

📝 HTML (`overflow.html`)

html

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="overflow.css">
</head>
<body>
  <div class="container">
    <p>This is a long paragraph that will not fit into the fixed height box. Let's see how the overflow property behaves when content goes beyond the height of the container.</p>
  </div>
</body>
</html>
```

🎨 CSS (`overflow.css`)

css

```
.container {
  width: 300px;
  height: 100px;
  border: 2px solid #333;
  padding: 10px;
  overflow: auto;
  background-color: #f8f8f8;
}
```

🔹 2. CSS Object-Fit

✅ When to Use:

- When you want to **control how images or videos are resized** inside a container.
- Prevents image distortion and ensures proper fitting.

🧠 Explanation:

The `object-fit` property defines how an image or video should be resized to fit its container.

css
```
object-fit: fill | contain | cover | none | scale-down;
```

- `fill`: stretches the image to fill the container (may distort).
- `contain`: fits the image completely without cropping (may leave empty space).
- `cover`: fills container by cropping the image (no empty space).
- `none`: keeps original size.
- `scale-down`: chooses smaller between `none` or `contain`.

📝 HTML (`objectfit.html`)

html
```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="objectfit.css">
</head>
<body>
  <h2>Object Fit: Contain</h2>
  <div class="image-container">
    <img src="https://via.placeholder.com/300x150" alt="demo image">
```

```
  </div>
</body>
</html>
```

🎨 CSS (`objectfit.css`)

css

```
.image-container {
  width: 200px;
  height: 150px;
  border: 2px solid #555;
  overflow: hidden;
}

.image-container img {
  width: 100%;
  height: 100%;
  object-fit: contain;
}
```

🔹 3. Object-Fit: Cover

✅ When to Use:

- Perfect for **profile pictures**, **hero sections**, or **image cards**.
- Keeps no empty space but may **crop** the image.

🧠 Explanation:

The `object-fit: cover` rule tells the image to fill the container entirely, cropping if necessary, while preserving aspect ratio.

📝 HTML (`objectcover.html`)

html

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="objectcover.css">
</head>
<body>
  <h2>Object Fit: Cover</h2>
  <div class="cover-container">
    <img src="https://via.placeholder.com/300x200" alt="cover image">
  </div>
</body>
</html>
```

🎨 CSS (`objectcover.css`)

css

```
.cover-container {
  width: 200px;
  height: 200px;
  border-radius: 10px;
  overflow: hidden;
  border: 2px solid #888;
}

.cover-container img {
  width: 100%;
  height: 100%;
  object-fit: cover;
}
```

✅ Summary

Topic	Use Case Example	Best For
Overflow	Scrollable divs, fixed modals	Manage excess content
Object-Fit	Image resizing for layout control	Clean, responsive media design
Object-Fit: Cover	Full-size image blocks like avatars	Full-frame visual impact